

3B5 Mechanics of Machines

Matlab Assignment Part 1

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Contents

Abstract	1
Pumpjack Dimensions	2
Graphical and Computational Solutions	3
Figure 1:	3
Figure 2:	3
Computing the Mechanics:	4
Approximating the System Mass and COM Positions	5
Global Manufacturing Locations	6
Small-Scale Model.....	6
Commented Final Code	7

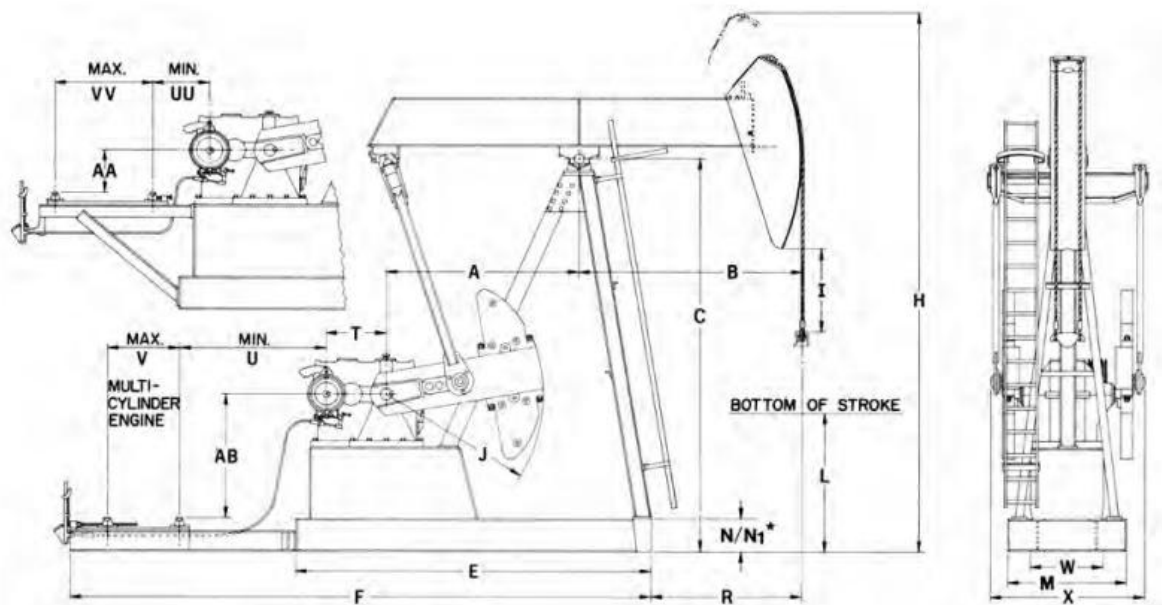
Abstract

For this assignment, we were tasked with designing a four-bar linkage mechanism with a real-world application. The design was to be simulated using MATLAB, and also constructed as a physical model. Out of the three available topics, I chose the first one which was, *'1. any real-world machine/mechanism which contains an application of a simple mechanism'*. My chosen mechanism was a pumpjack system, also known as a 'Nodding Donkey' or 'Horsehead Pump', from a traditional oil well. This mechanism uses a four-bar linkage to operate in a crank-rocker fashion, which is used to mechanically lift oil out of a well in a case where there is not enough bottom hole pressure for the liquid to flow all the way to the surface. The system typically uses an electric motor to drive a crank shaft which is connected to a counterweight. The crank shaft is linked to the walking beam via the pitman arm (coupler), which causes the walking beam to oscillate vertically, driving the polished rod and sucker rod into the hole. This in turn creates pressure which helps extract oil and gas from the earth.

This report includes an explanation of what a pumpjack is, how a 4-bar simulation was performed, the calculations controlling the simulation, manufacturing locations, details about the small-scale model, and the final commented code.

Pumpjack Dimensions

Globally, the dimensions for each pumpjack obviously vary, but in order to try to simulate a mechanism that is as accurate as possible I used data from Lufkin Industries. Lufkin is one of the leading manufacturers of pumpjacks and other mechanical devices that are used in the oil and gas industry. I used the data from the ‘Lufkin General Catalog 2006’ for the pumpjack dimensions. The datasheets can be accessed using the following link: [Lufkin General Catalog 2006](#).

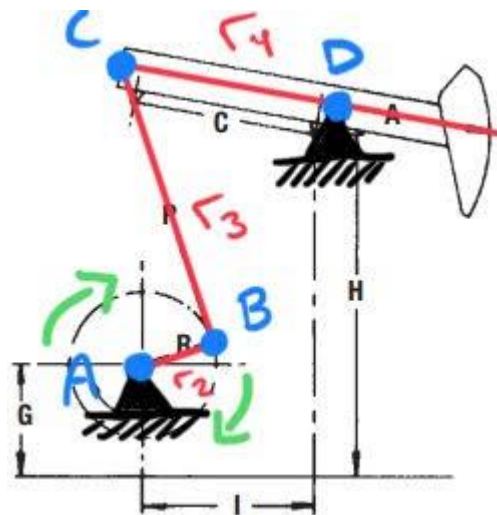


The above image shows a schematic of a LUFKIN Industries pumpjack with labelled dimensions. From the data sheet, I took the dimensions from the ‘C-912D-427-144’ model. This is a medium sized system. The dimensions as per the table are:

$$r_2 = R_1 = 47 \text{ inches} = \underline{1.1938 \text{ m}}$$

$$r_3 = P = 148.50 \text{ inches} = \underline{3.7719 \text{ m}}$$

$$r_4 = A + C = 180 + 120.03 \\ = 300.03 \text{ inches} = \underline{7.62 \text{ m}}$$



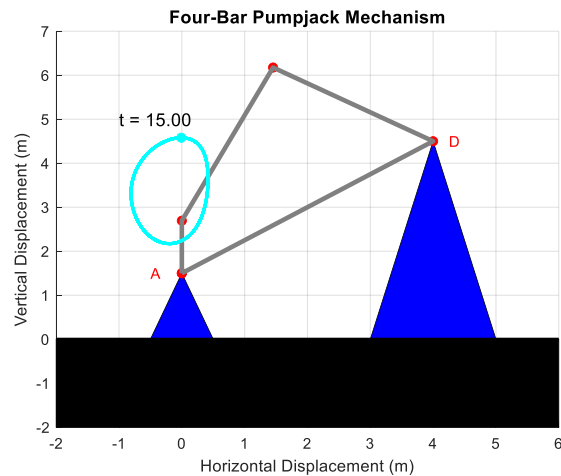
Unit Size	A	C	I	P	H	G	R1, R2, R3, R4	S.U.*	T.F. @ 90°/Stroke Length
C-912D-427-144	180	120.03	120	148.50	262.00	111.00	47. 41. 35	-650	68.82/144

These dimensions were used in the MATLAB simulation, where 1 square on the grid was equal to 1m^2 .

Graphical and Computational Solutions

Figure 1:

The graph to the right shows a simplified model of the pumpjack, generated in MATLAB. The system is broken down into a 4-bar linkage consisting of a crank, coupler, rocker, and frame. In the case of a pumpjack, the rocker is referred to as the walking beam. The two blue triangular stilts were added to the diagram, so the system better resembled a pumpjack. The triangles represent the fixed points that connect the links to the grounded joints A and D. The lengths of each of the links were set to the chosen dimensions from the ‘Lufkin General Catalog 2006’, as shown above. The crank speed was determined from the source, [One Step Power](#), and was found to be up to 20 strokes per minute.



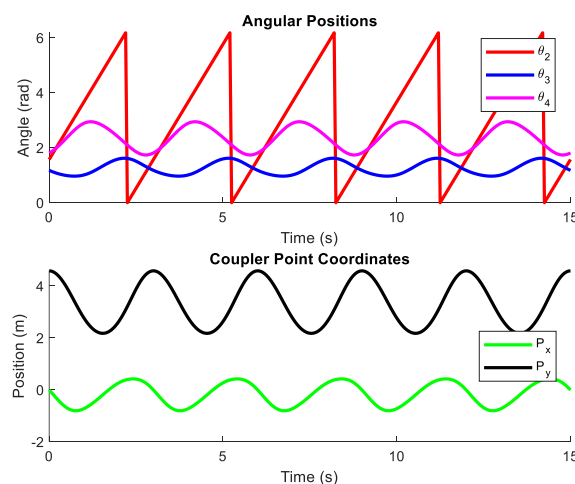
$$20 \text{ strokes per min} = 20 \text{ RPM}$$

$$\omega = \text{RPM} \times \frac{2\pi}{60} = 20 \times \frac{2\pi}{60} = \frac{20\pi}{30} \text{ rads}^{-1}$$

This works out to be approximately 2.09 rad/s. This crank speed was used in the simulation. The loop was set for 15 second, so the crank rotated 10π radians, or 5 whole revolutions/strokes. The rotary motion of the crank turns the coupler which pulls the rocker. Due to the positions of the fixed joints, in this case the rocker oscillates vertically. Ideally in a practical application, the rocker (walking beam) would extend beyond the point D and have a vertically positioned ‘polished rod’, which would be pulled up and down into the well. However, this simulation was just to show the mechanics of the 4-bar linkage system within the pumpjack.

Figure 2:

The figure to the right shows two graphical representation of the angular positions (top) and the coupler point coordinates (bottom) w.r.t time. For the top graph, the red line represents the crank angle, the blue line represents the coupler angle, and the magenta line represents the rocker angle. The crank angle is a sawtooth pattern because once the crank completes 1 full revolution ($2\pi \approx 6.28$ rad), the line returns to zero. Otherwise, it would form a continuous straight line for as long as the crank is



in constant motion. The bottom graph shows the coupler point x and y coordinates w.r.t time. This point was defined as point P in the simulation. It was placed halfway along the coupler link (r_3) at an angle of 0.4 rad relative to the link. This point was used to track the coupler movement as the system was in motion. A tracer function was used to map out its total displacement once the system had come to a halt after 15 seconds.

The result of these 2 graphs was made possible due to the system being a closed loop. Mathematically this can be expressed as:

$$\vec{r}_{AB} + \vec{r}_{BC} = \vec{r}_{CD} + \vec{r}_{DA} \quad \text{or} \quad \vec{r}_2 + \vec{r}_3 = \vec{r}_4 + \vec{r}_1$$

Computing the Mechanics:

To start, the 2 fixed points were defined:

$$A = [0, 1.5], \quad B = [4, 4.5]$$

Then, the link lengths (r_x) were defined:

$$\vec{r}_1 = \sqrt{(D_x - A_x)^2 + (D_y - A_y)^2}, \quad \vec{r}_2 = [AB], \quad \vec{r}_3 = [BC], \quad \vec{r}_4 = [CD]$$

An initial crank angle and angular velocity was defined:

$$\theta_2 = \frac{\pi}{2} \text{ rad}, \quad \dot{\theta}_2 = \frac{20\pi}{30} \text{ rads}^{-1}$$

Using our initial conditions, we solve for the coordinates of point B:

$$B_x = A_x + r_2 \cos(\theta_2), \quad B_y = A_y + r_2 \sin(\theta_2)$$

The construction vector was calculated using points B and D:

$$S_x = D_x - B_x, \quad S_y = D_y - B_y$$

$$\vec{s} = \sqrt{S_x^2 + S_y^2}, \quad \theta_s = \tan^{-1}\left(\frac{S_y}{S_x}\right)$$

From here, we calculated cosine of the coupler (3) and walking beam (4) angles:

$$\cos(\theta_3) = \left(\frac{\vec{s}^2 + \vec{r}_3^2 - \vec{r}_4^2}{2\vec{s}\vec{r}_3} \right), \quad \cos(\theta_4) = \left(\frac{-\vec{s}^2 + \vec{r}_3^2 + \vec{r}_4^2}{2\vec{r}_3\vec{r}_4} \right)$$

These calculations were executed in a loop using an iterator denoted as 'i'.

Approximating the System Mass and COM Positions

Assuming that the entire structure is made of steel, we can calculate a rough approximation of the total system mass and centre of mass of the pumpjack. The walking beam and main frame are often made from ASTM A36 steel, which has a density of 7.85 g/cm^3 . The pitman arm (coupler), the crank, and the counterweights use high-strength, low alloy steel such as ASTM A514, which has a density of 7.80 g/cm^3 . Obviously, the system has different widths at different points, so for ease I will let each bar have a width and thickness of $20 \times 20 \text{ cm}$. Using this assumption, the volume and mass of each component can be calculated:

Part	Density (g/cm^3)	Length (m)	Volume (m^3)	Mass (kg)
Crank	7.80	1.1938	0.047752	372.47
Coupler	7.80	3.7719	0.150876	1176.83
Walking Beam	7.85	7.62	0.3048	2392.68

Therefore, total mass of the moving components for the simplified 4-bar system is 3941.98 kg. To find the centre of mass for the whole system, we find the COM for each link. We can assume that each link has a uniformly distributed mass.

Crank [AB] = [(0, 1.5), (B_x , B_y)]:

$$\begin{aligned}
 B_x &= A_x + 1.1938 \cos(90^\circ) = 0 + 1.1938(0) = 0 \\
 B_y &= A_y + 1.1938 \sin(90^\circ) = 1.5 + 1.1938(1) = 2.6938 \\
 COM_{crank} &= \left(\frac{0 + 0}{2}, \frac{1.5 + 2.6938}{2} \right) = (0, 2.0969)
 \end{aligned}$$

Coupler [BC] = [(0, 2.0969), (C_x , C_y)]:

$$\begin{aligned}
 C_x &= B_x + 3.7719 \cos(0^\circ) = 0 + 3.7719(1) = 3.7719 \\
 C_y &= B_y + 3.7719 \sin(0^\circ) = 2.6938 + 3.7719(0) = 2.6938 \\
 COM_{coupler} &= \left(\frac{0 + 3.7719}{2}, \frac{2.6938 + 2.6938}{2} \right) = (1.88595, 2.6938)
 \end{aligned}$$

Walking Beam [CD] = [(1.88595, 2.6938), (4, 4.5)]:

$$COM_{walking\ beam} = \left(\frac{1.88595 + 4}{2}, \frac{2.6938 + 4.5}{2} \right) = (3.88595, 3.5969)$$

Using the individual COM values, we can use the weighted average formula to find the system COM:

$$COM_x = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}, \quad COM_y = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$

The result from this was: $COM_{system} = (2.92, 3.19) \text{ meters}$

The above calculations were also implemented into the MATLAB script and printed in the terminal shown on the right:

```
Command Window
>> FOC_3B5_Assignment1_Part1
Center of Mass (X, Y): (2.92, 3.19) meters
Total Mass: 3941.98 kg
fx >>
```

Global Manufacturing Locations

Pumpjacks are manufactured all over the world. There are a lot of different companies who manufacture pumpjacks and other equipment for the energy industry. Schlumberger Limited are headquartered in Houston, Texas. Halliburton Company are also headquartered in Houston, Texas with manufacturing facilities in the USA, Malaysia, Singapore and the UK. LUFKIN Industries are headquartered in Missouri City, Texas and produce downhole pumps in Canada and pumping units in Asia and Europe. Sanjack Petro also manufacture pumping units and are based in Dongying, China. There are also other manufacturers in the Middle East, Europe, and other parts of the USA and Asia. Houston is often regarded to as the “Energy Capital of the World”, with companies such as Exxonmobil, Marathon Oil, and Shell having offices in the city.

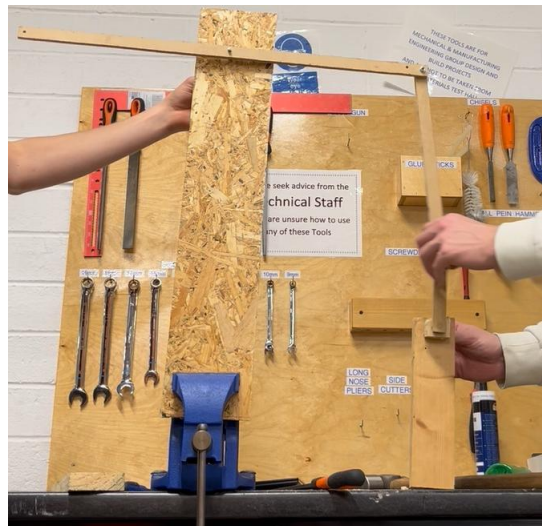
Small-Scale Model

To physically simulate the pumpjack mechanism, I built a small-scale model using plywood, nuts, bolts, and washers. The crank was turned using my hand. Below is a series of photos showing the walking beam (rocker) displacement when the crank is positioned at different angles.

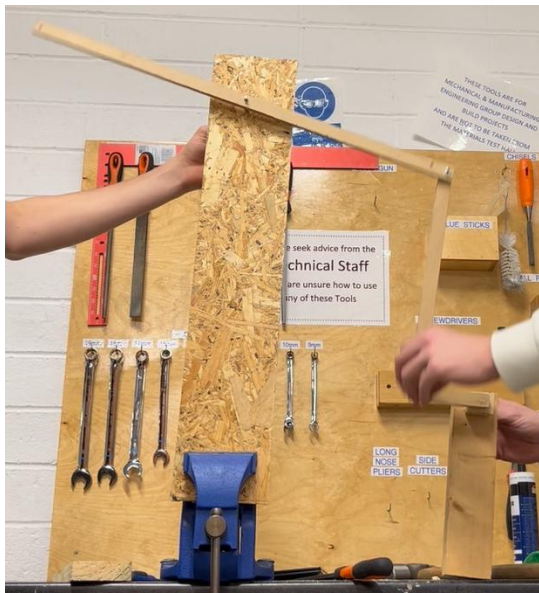
Crank Angle at 0 rad:



Crank Angle at $\pi/2$ rad:



Crank angle at π rad:



Crank angle at $3\pi/4$ rad:



Commented Final Code

```
% Mechanical Simulation for an Oil Well Pump Jack
% Finn O'Connor 22336740

close all; clear;

%% Define mechanism dimensions and fixed joint positions
Ax = 0; Ay = 1.5; % A coordinates
Dx = 4; Dy = 4.5; % D coordinates
r1 = sqrt((Dx - Ax)^2 + (Dy - Ay)^2); % Ground link length
r2 = 1.1938; % Crank length
r3 = 3.7719; % Coupler length
r4 = 3.048762; % Rocker length

BP = r3 / 2; % Coupler point P distance from B
theta_5 = 0.4; % Offset angle for P

%% Define crank initial conditions and simulation parameters
theta2_initial = pi/2; % Initial crank angle
theta2d = (20 * pi) / 30; % Calculated crank angular speed
t_final = 15;
dt = 0.05;
t = 0 : dt : t_final;
N = length(t);

%% Initialise arrays
time = zeros(1, N);
theta2 = zeros(1, N);
theta3 = zeros(1, N);
theta4 = zeros(1, N);
Bx = zeros(1, N); By = zeros(1, N);
Cx = zeros(1, N); Cy = zeros(1, N);
Px = zeros(1, N); Py = zeros(1, N);
s = zeros(1, N);
thetas = zeros(1, N);
```

```

%% Compute center of mass (COM) of the four-bar linkage
% Material densities in kg/m^3
density_crank = 7800;
density_coupler = 7800;
density_walking_beam = 7850;

% Cross-sectional dimensions (m)
width = 0.2;
thickness = 0.2;

% Compute volumes (m^3)
volume_crank = width * thickness * r2;
volume_coupler = width * thickness * r3;
volume_walking_beam = width * thickness * 7.62;

% Compute masses (kg)
mass_crank = density_crank * volume_crank;
mass_coupler = density_coupler * volume_coupler;
mass_walking_beam = density_walking_beam * volume_walking_beam;

total_mass = mass_crank + mass_coupler + mass_walking_beam;

% Compute center positions
Bx = Ax + r2 * cos(pi / 2);
By = Ay + r2 * sin(pi / 2);
Cx = Bx + r3 * cos(theta);
Cy = By + r3 * sin(theta);

% Center of mass for each component
center_crank = [(Ax + Bx) / 2, (Ay + By) / 2];
center_coupler = [(Bx + Cx) / 2, (By + Cy) / 2];
center_walking_beam = [(Cx + Dx) / 2, (Cy + Dy) / 2];

% Compute total center of mass of system
COM_x = (mass_crank * center_crank(1) + mass_coupler * center_coupler(1) +
mass_walking_beam * center_walking_beam(1)) / total_mass;
COM_y = (mass_crank * center_crank(2) + mass_coupler * center_coupler(2) +
mass_walking_beam * center_walking_beam(2)) / total_mass;

fprintf('Center of Mass (X, Y): (%.2f, %.2f) meters\n', COM_x, COM_y);
fprintf('Total Mass: %.2f kg\n', total_mass);

%% Figure 1: Animation
figure(1); hold on; grid on;
title('Four-Bar Pumpjack Mechanism', 'FontSize', 12);
xlabel('Horizontal Displacement (m)'); ylabel('Vertical Displacement (m)');
xlim([-2, 6]);
ylim([-2, 7]);
text(-0.5, 1.5, 'A', 'Color', 'r');
text(4.25, 4.5, 'D', 'Color', 'r');
fill([-2 6 6 -2], [0 0 -2 -2], 'k-'); % Ground
fill([-0.5 0.5 0], [0 0 1.5], 'b-'); % Small fixed support
fill([3 5 4], [0 0 4.5], 'b-'); % Large fixed elevated support
plot([-2, 6], [0, 0], 'k-', 'LineWidth', 2); % Ground line

h1 = plot(0, 0, 'ro', 'MarkerFaceColor', 'r');
h2 = plot(0, 0, 'Color', [0.5 0.5 0.5], 'LineWidth', 3);
h3 = text(-1, 5, 't = 0', 'FontSize', 12);

```



```

h4 = plot(0, 0, 'co', 'MarkerFaceColor','c','MarkerSize',6);
coupler_trace_x = []; coupler_trace_y = []; % Initialises dynamic arrays

%% Main simulation loop
i = 0;
for t = 0 : dt : t_final
    i = i + 1;
    time(i) = t;

    % Crank angle calculation (modulus used to keep positive)
    theta2(i) = mod(theta2_initial + theta2d*t, 2*pi);

    % X and Y coordinate calculation for point B
    Bx(i) = Ax + r2 * cos(theta2(i));
    By(i) = Ay + r2 * sin(theta2(i));

    % Construction vector s ([B, D]) calculation
    sx = Dx - Bx(i); sy = Dy - By(i);
    s(i) = sqrt(sx^2 + sy^2);
    thetas(i) = atan2(sy, sx); % atan2() calculates the quadrants

    % Coupler angle w.r.t x-axis
    cos_theta3 = (s(i)^2 + r3^2 - r4^2) / (2 * s(i) * r3);

    % Walking beam angle
    cos_theta4 = (-s(i)^2 + r3^2 + r4^2) / (2 * r3 * r4);

    % If statement to ensure that complex values are ignored
    if abs(cos_theta3) <= 1 && abs(cos_theta4) <= 1
        theta3(i) = thetas(i) + acos(cos_theta3);
        theta4(i) = thetas(i) + acos(cos_theta4);
        Cx(i) = Bx(i) + r3 * cos(theta3(i));
        Cy(i) = By(i) + r3 * sin(theta3(i));
        Px(i) = Bx(i) + BP * cos(theta3(i) + theta_5);
        Py(i) = By(i) + BP * sin(theta3(i) + theta_5);
    else
        theta3(i) = NaN; theta4(i) = NaN;
        Cx(i) = NaN; Cy(i) = NaN;
        Px(i) = NaN; Py(i) = NaN;
    end

    set(h1, 'XData', [Ax, Bx(i), Cx(i), Dx, Ax], 'YData', [Ay, By(i), Cy(i), Dy, Ay]);
    set(h2, 'XData', [Ax, Bx(i), Cx(i), Dx, Ax], 'YData', [Ay, By(i), Cy(i), Dy, Ay]);
    set(h4, 'XData', Px(i), 'YData', Py(i));
    set(h3, 'String', ['t = ' num2str(t, '%.2f')]);

    % Tracer used to track the displacement of the couple point P
    coupler_trace_x(end+1) = Px(i);
    coupler_trace_y(end+1) = Py(i);
    drawnow; pause(0.01);
end

plot(coupler_trace_x, coupler_trace_y, 'c-', 'LineWidth',2);

%% Figure 2: Time-History Plots
% clf to clear plot; gca to get current axis
figure(2); clf; set(gca, 'FontSize',14); grid on;

```

```

subplot(2,1,1); hold on;
plot(time, theta2, 'r-', 'LineWidth',2);
plot(time, theta3, 'b-', 'LineWidth',2);
plot(time, theta4, 'm-', 'LineWidth',2);
legend('\theta_2','\theta_3','\theta_4','Location','best');
xlabel('Time (s)'); ylabel('Angle (rad)');
title('Angular Positions');

subplot(2,1,2); hold on;
plot(time, Px, 'g-', 'LineWidth',2);
plot(time, Py, 'k-', 'LineWidth',2);
legend('P_x','P_y','Location','best');
xlabel('Time (s)'); ylabel('Position (m)');
title('Coupler Point Coordinates');

```